Abir Sarkar

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Education

Cornell University 2025–2029 (Expected)
Ph.D. in Statistics and Data Science *Ithaca, NY*
Advisor: Prof. Martin T. Wells

Indian Statistical Institute 2021–2023
Master of Statistics, specialization in Finance *Kolkata, India*

Indian Statistical Institute 2018–2021
Bachelor of Statistics *Kolkata, India*

Teaching Experience at Cornell University

ECON 2801: Game Theory—For Economics, Politics, Knowledge, and Rationality Fall 2025
Grader, Cornell University *Instructor: Prof. Kaushik Basu*

STSCI 4981: Sports Analytics in Practice Spring 2026
Teaching Assistant and Grader *Instructor: Prof. Martin T. Wells*

STSCI 5080: Probability Models and Inference Fall 2026
Teaching Assistant and Grader *Instructor: Prof. Kengo Kato*

Publications

Is There an AI Bubble? Robust Date-Stamping for Periods of Exuberance
Accepted in Frontiers of Mathematical Finance
Abir Sarkar and Martin T. Wells

- We developed a stochastic-volatility-robust ADF framework for date-stamping asset-price bubbles under persistent and time-varying volatility, deriving nuisance-parameter-free asymmetric calibration rules for bubble origination and collapse.
- We established consistent date-stamping and a practically implementable procedure for real-time monitoring. Our method demonstrates improved power and dating accuracy relative to conventional homoskedastic PWY tests. We applied the method to the Magnificent Seven and major semiconductor stocks, finding substantial cross-sectional heterogeneity in price exuberance.

Asymptotic Breakdown Point Analysis for a General Class of Minimum Divergence Estimators
Accepted in Bernoulli
Subhrajyoty Roy, **Abir Sarkar**, Abhik Ghosh, and Ayanendranath Basu

- We investigated the global robustness of minimum S-divergence estimators under general parametric models and developed asymptotic breakdown-point theory for a broad class of divergence-based estimators that serve as robust alternatives to maximum likelihood.
- The resulting dimension-free robustness guarantees provide theoretical support for the use of minimum-divergence procedures in high-dimensional statistical inference.

Submitted Articles

Adapt or Forget: Provable Tradeoffs between Adam and SGD in Nonstationary Optimization
Preprint, Submitted to NeurIPS
Sharan Sahu*, **Abir Sarkar***, Cameron J. Hogan*, and Martin T. Wells

- We developed a finite-time analysis of Adam under nonstationary stochastic objectives and identified an explicit noise–drift tradeoff: adaptive averaging and preconditioning are beneficial in noise-dominated regimes, while stale momentum and preconditioner lag can make SGD preferable when objective drift dominates.
- Our experiments across convex and nonconvex problems support this tradeoff, showing that Adam performs better under high noise and low drift, whereas SGD is often more robust under high drift and low noise.

Double Local-to-Unity: Inference under Nearly Nonstationary Volatility

Preprint, Submitted to Journal of Econometrics

Abir Sarkar and Martin T. Wells

- We developed a moderate-deviation limit theory for autoregressive models in which both the autoregressive root and the stochastic-volatility persistence parameter approach unity at distinct rates.
- We demonstrated through applications to historical bubbles, commodities, cryptocurrencies, and global real-estate markets that the proposed procedure is more robust to volatility spikes and spurious explosive signals than standard homoskedastic methods.

Empirical Bayes Predictive Density Estimation under Covariate Shift in Large Imbalanced Linear Mixed Models

Preprint, Submitted to JASA

Abir Sarkar, Gourab Mukherjee, and Keisuke Yano

- We studied empirical Bayes predictive-density estimation in high-dimensional linear mixed models with many units, severe data imbalance, limited replication, and covariate shift, and developed a calibration framework for optimally tuning predictive densities generated from flexible prior classes under Kullback–Leibler risk.
- We introduced a data-fission risk estimator that exploits covariate exchangeability, established convergence rates under data scarcity, and proved decision-theoretic optimality of the resulting empirical Bayes predictor through a novel analysis of data reuse and predictive heat-equation representations.

Industry Experience

Capital One Jun 2023–Jul 2025

Senior Associate, Data Scientist, Model Validation *Bangalore, India*

- Validated Commercial and Industrial, Commercial Real Estate, and Structured Finance probability-of-default and loss-given-default models covering more than $140 billion in portfolio exposure.
- Led independent validation of a $7 billion Municipal LGD model and approximately $20 billion in Healthcare C&I and Healthcare CRE portfolios.
- Conducted conceptual-soundness reviews, benchmarking, sensitivity analysis, independent model replication, prototype testing, implementation testing, and deployment validation.
- Contributed to retail-fraud modeling, consumer-loss forecasting, deposit modeling, cannibalization models, and CCAR and CECL stress-testing initiatives.
- Supported internal governance and regulatory reviews, completed two audit examinations with no findings.

Internships

JPMorgan Chase & Co. Summer 2022

Market Risk Analyst Intern *Mumbai, India*

- Evaluated the performance of trading and risk models used in foreign exchange and commodities businesses within the Corporate and Investment Bank.
- Developed a quantitative framework for rating models as part of ongoing performance monitoring.

Big Data Summer Institute, University of Michigan Summer 2021

Research Intern and Team Lead *Ann Arbor, MI*

- Led a team analyzing Michigan Medicine data and developed missing-data imputation pipelines using regression, multiple imputation, regression trees, random forests, and XGBoost.
- Used the imputed clinical data to construct and evaluate statistical models for predicting hospitalization outcomes.

Selected Other Projects

Machine-Learning Methods for Fraud Detection

Indian Statistical Institute, under Prof. Soumendu Sundar Mukherjee

Developed and compared ensemble-learning methods for transaction-level fraud detection, including feature selection, hyperparameter tuning, and SHAP-based interpretation.

Robust Testing for Removing Unwanted Variation

Indian Statistical Institute, Master's Project, under Prof. Abhik Ghosh

Developed robust testing procedures for detecting differentially expressed genes in the presence of unwanted variation.

Conference Presentations

- **Is There an AI Bubble? Robust Date-Stamping for Periods of Exuberance**
  - The 39th New England Statistics Symposium (NESS), *From Data to Discovery: Statistical Insight in the Age of AI*, University of Connecticut, Storrs, CT (2026).
- **Empirical Bayes Predictive Density Estimation under Covariate Shift in Large Imbalanced Linear Mixed Models**
  - Annual Conference of the International Indian Statistical Association, Kochi, India (2024).
  - International Conference on Statistics and Data Science, Seville, Spain (2025).

Awards and Achievements

- **Director's Award**, Capital One, for contributions to the validation of Commercial C&I and Municipal default models.
- **Statistically Significant Award**, Capital One, for exceptional performance across multiple projects under demanding timelines.
- **INSPIRE Scholarship**, Department of Science and Technology, Government of India, 2018–2023.
- Secured **1st rank** in the **Jagadis Bose National Science Talent Search (JBNSTS)**, 2018.
- Ranked **3rd** in both the B.Stat. (2018–2021) and M.Stat. (2021–2023) cohorts at the Indian Statistical Institute.
- Ranked **4th** and **5th** in the **Simon Marais Mathematics Competition** in 2018 and 2019, respectively.

Technical Skills

Programming: Python, R, SQL, C++, LaTeX, Git
Language: English, Hindi, Bengali